

		particles, unrelated to nucleus of the gold atom.
30	A	Actual initial count rate of the radioactive sample, $C_0 = 250 - 50 = 200 \text{ min}^{-1}$ Actual count rate of sample at $t = 50 \text{ s}$, $C_{50} = 150 - 50 = 100 \text{ min}^{-1}$ Actual count rate of sample at $t = 100 \text{ s}$, $C_{100} = 100 - 50 = 50 \text{ min}^{-1}$ Actual count rate of sample at $t = 150 \text{ s}$, $C_{150} = 75 - 50 = 25 \text{ min}^{-1}$ By observation, the half-life of the radioactive sample $t_{\frac{1}{2}} = 50 \text{ s}$